

SIMPHIWE KUBHEKA

Electrical & Computer Engineering — Embedded Systems · Firmware · Real-Time Software



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01 PROFILE

Final-year Electrical and Computer Engineering student at the University of Cape Town, focused on embedded systems, firmware development, real-time software, and cyber-physical systems. Experience spans register-level STM32 development, embedded kernel design, low-power data acquisition, autonomous control, and secure peer-to-peer communication research — building reliable systems at the intersection of hardware and software.

02 EDUCATION

University of Cape Town BSc Eng, Electrical & Computer Engineering GPA 75% Dean's Merit List 2024 & 2025	2023 — PRESENT	Emhlwaneni High School National Senior Certificate — 8 distinctions Top 20 Matric, KwaZulu-Natal NSTF Scholarship — Maths & Physics	2018 — 2022
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03 SELECTED ENGINEERING PROJECTS

ARM Cortex-M7 Real-Time Kernel A lightweight preemptive RTOS kernel built on static allocation. - Priority-based scheduling with 32 priority levels, O(1) ready lists, and dynamic-tick timekeeping. - Mutexes with transitive priority inheritance, counting semaphores, queues, task notifications, and ISR-safe variants. - MPU-based stack-overflow guard and optional C++ RAII wrappers over the C kernel API.	STM32H755 · C / C++
STM32F446 Bare-Metal Driver Library A peripheral library written without STM32 HAL/LL dependencies. - RCC, GPIO, EXTI, NVIC, SysTick, PWR, ADC, DMA, and modular timer drivers covering PWM, encoder, input capture, synchronisation, and motor control. - Custom startup code, linker configuration, memory layout, Makefile build system, and an OpenOCD/GDB debug workflow.	REGISTER-LEVEL C
Secure Communication in a Scalable Peer-to-Peer Embedded Architecture Investigating secure communication for scalable, resource-constrained embedded systems. - Literature research and evaluation of communication and security approaches for decentralised embedded architectures. - ESP32-based prototype for secure peer-to-peer voice communication, with emphasis on scalability, resource constraints, and real-time performance.	FINAL-YEAR RESEARCH
Low-Power Optical Data Acquisition System Firmware for a battery-operated optical data logger designed for autonomous operation. - State-machine-driven acquisition pipeline with timer-controlled differential sampling, ambient-light cancellation, temperature compensation, CRC-32, and SD-card logging. - RTC scheduling, Stop 2 power management, watchdog recovery, centralised fault handling, and persistent fault history across watchdog resets.	STM32L476RG

03 SELECTED PROJECTS – CONTINUED

Autonomous Micromouse Navigation

MATLAB · SIMULINK · STATEFLOW

An autonomous maze-navigation and motion-control system.

- Depth-first search with backtracking for maze exploration and breadth-first search for shortest-path computation.
- PID distance/heading control and sensor fusion using wheel encoders and gyroscope measurements, with time-of-flight wall following.

04 EXPERIENCE

Embedded Systems Tutor – University of Cape Town

2026 – PRESENT

- Tutor for EEE2046S (Embedded Systems I) and EEE3096S (Embedded Systems II), supporting students with embedded programming, microcontroller concepts, and practical engineering problem-solving.

Electronics Tutor – University of Cape Town

2025 – PRESENT

- Support students in circuit analysis and electronics concepts, reinforcing theory through practical problem-solving.

Residence Tutor – Leo Marquard Residence / Obz Square

2024 – PRESENT

- Provide academic mentorship and structured support to students in residence.

05 TECHNICAL SKILLS

PROGRAMMING	C	C++	Python	MATLAB					
EMBEDDED	ARM Cortex-M	STM32	CMSIS	Bare-Metal Firmware	RTOS	Interrupts	DMA		
	Timers	ADC	UART	SPI	I ² C				
CONTROL & MODELLING	Simulink	Stateflow	PID Control						
TOOLS	Git	GitHub	OpenOCD	GDB	STM32CubeIDE	Make			
SYSTEMS	Data Acquisition	Low-Power Design	Embedded Communication	Signal Processing					

06 ADDITIONAL TECHNICAL EXPOSURE

5G / Wi-Fi 6 Unlicensed-Band Listen-Before-Talk

VACATION WORK

- Exposure to OpenAirInterface, USRP B210 software-defined radios, Docker-based development, and the implementation and evaluation of Listen-Before-Talk mechanisms.